M3 · Pattern recognition receptors & signalling

Innate immunity lecture pack · EAACI/UEMS Knowledge Exam preparation

After this module you can

- Define PAMPs and DAMPs and give typical examples.
- Place each PRR family in its cell compartment: cell surface, endosome, cytosol or plasma.
- Name the ligand and location of each human Toll-like receptor, including TLR9 and CpG DNA.
- Follow TLR4 signalling through MyD88 to NF-κB and through TRIF to IRF3 and IFN-β.
- Explain the two-signal NLRP3 inflammasome, pyroptosis and the IL-1-driven autoinflammatory syndromes.
- Describe cytosolic RNA and DNA sensing (RIG-I, MDA5, MAVS, cGAS–STING, AIM2).
- Match inborn errors of PRR signalling to their clinical phenotype.

AIM2	absent in melanoma-2
ASC	apoptosis-associated speck-like protein containing a CARD
CAPS	cryopyrin-associated periodic syndromes
CARD	caspase recruitment domain
cGAS / cGAMP	cyclic GMP-AMP synthase / cyclic GMP-AMP
CLR	C-type lectin-like receptor
CRP	C-reactive protein
DAMP	damage-associated molecular pattern
DC / pDC	dendritic cell / plasmacytoid dendritic cell
FPR1	formyl peptide receptor 1
HSE	herpes simplex encephalitis
HMGB1	high-mobility group box 1
IFN	interferon
IKK	inhibitor of NF-κB kinase
IRAK	IL-1 receptor-associated kinase
IRF	interferon regulatory factor
LPS	lipopolysaccharide
LTA	lipoteichoic acid
MAVS	mitochondrial antiviral-signalling protein
MBL	mannose-binding lectin
MDA5	melanoma differentiation-associated gene 5
MyD88	myeloid differentiation factor 88
MAL	MyD88-adaptor-like
NEMO	NF-κB essential modulator (IKKγ)
NF-κB	nuclear factor kappa B
NLR	NOD-like receptor
NOD	nucleotide oligomerization domain
PAMP	pathogen-associated molecular pattern
PRR	pattern recognition receptor
RIG-I / RLR	retinoic acid-inducible gene I / RIG-like receptor

SAP	serum amyloid P
STING	stimulator of IFN genes
TBK1	TANK-binding kinase 1
TIR	Toll/IL-1 receptor domain
TLR	Toll-like receptor
TRAF	TNF receptor-associated factor
TRIF / TRAM	TIR domain-containing adaptor inducing IFN-β / TRIF-related adaptor molecule

1. What the innate system recognizes: PAMPs and DAMPs

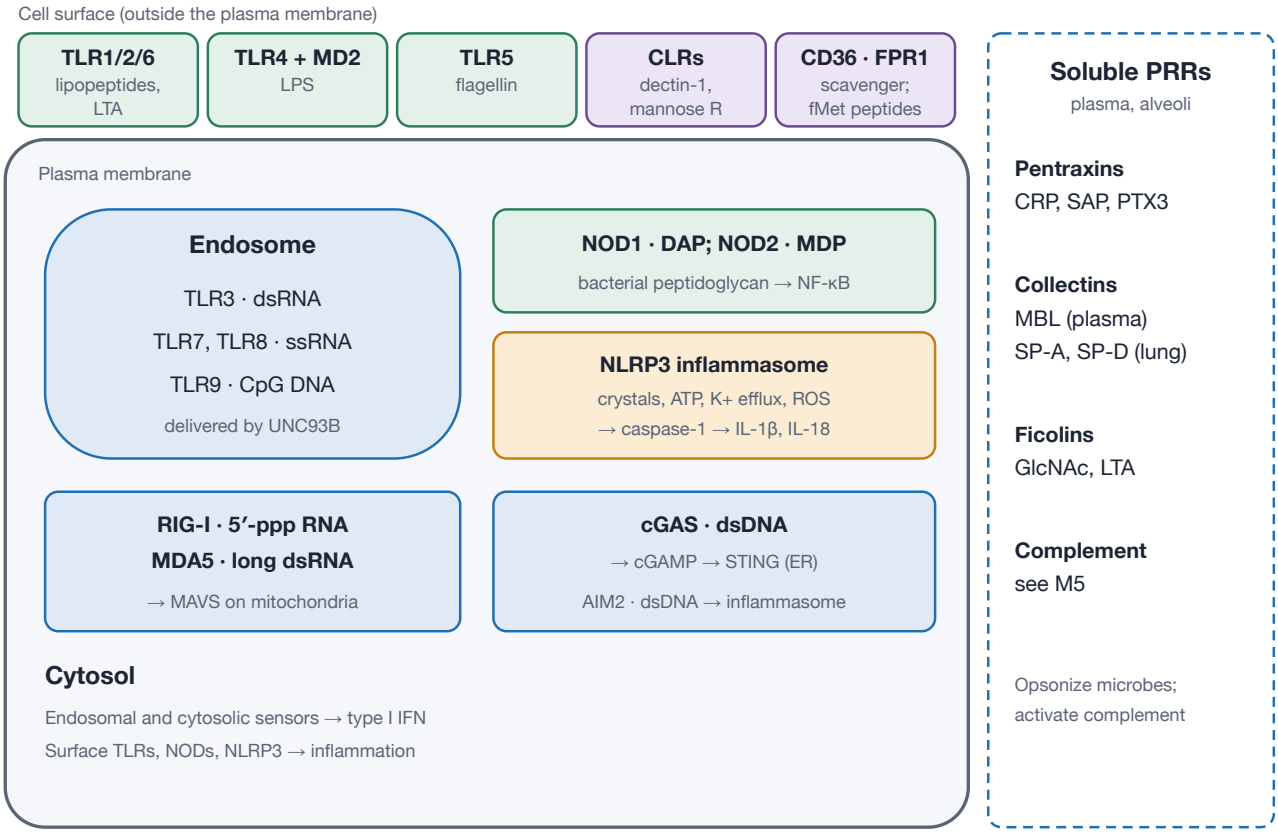
Innate cells do not recognize single microbial species. They recognize molecular patterns that whole classes of microbes share. These are pathogen-associated molecular patterns (PAMPs). Abbas p.63 Typical PAMPs are double-stranded RNA of replicating viruses, unmethylated CpG DNA of bacteria, proteins that start with N-formylmethionine, lipopolysaccharide (LPS), lipoteichoic acid (LTA) and oligosaccharides with terminal mannose. Abbas p.63

Damaged and dying host cells release endogenous molecules that also start innate responses. These are damage-associated molecular patterns (DAMPs). DAMPs can signal sterile injury, for example burns, trauma or loss of blood supply. Abbas p.63 Examples are heat shock proteins, monosodium urate crystals, extracellular ATP and formylated mitochondrial peptides, extracellular HMGB1, histones and cytoplasmic dsDNA. Abbas p.65

The cellular receptors for PAMPs and DAMPs are pattern recognition receptors (PRRs). They sit on the cell surface, in endosomes (phagocytic vesicles) and in the cytosol. Soluble recognition molecules circulate in blood and secretions. Abbas p.64

> **Analogy**
>
> Think of PRRs as smoke detectors in different rooms of a house. A detector on the outside wall (cell surface) sees bacterial walls. A detector in the waste room (endosome) sees nucleic acids from digested microbes. A detector in the living room (cytosol) sees invaders that have already broken in.

2. The PRR map



D7. PRR location map. Surface TLRs and lectin-type receptors see the outside of microbes. Endosomal TLRs see nucleic acids of ingested microbes. Cytosolic sensors see microbes or damage inside the cell; soluble PRRs act in plasma and alveolar fluid. Abbas p.64–89

Family	Location	Examples	Main ligands	Main output
TLRs	Plasma membrane; endosome	TLR1–TLR10	LPS, lipopeptides, flagellin, viral and bacterial nucleic acids	NF-κB (inflammation); IRF (type I IFN)
NLRs	Cytosol	NOD1, NOD2; NLRP3	Peptidoglycan fragments; crystals, ATP, K+ efflux	NF-κB; inflammasome → IL-1β, IL-18
RLRs	Cytosol	RIG-I, MDA5	Viral RNA	Type I IFN
Cytosolic DNA sensors	Cytosol	cGAS–STING; AIM2	Microbial or damaged-self dsDNA	Type I IFN; inflammasome
CLRs	Plasma membrane of phagocytes and DCs	Mannose receptor, dectin-1	Terminal mannose/fucose; β-glucan	Phagocytosis; NF-κB, Th17
Scavenger receptors	Plasma membrane of macrophages	SR-A, CD36	Diverse microbial molecules	Phagocytosis; TLR2/6 co-receptor
Formyl peptide receptor	Plasma membrane of leukocytes	FPR1	N-formylmethionyl peptides	Chemotaxis
Soluble PRRs	Plasma; alveoli	CRP, SAP, PTX3; MBL, SP-A, SP-D; ficolins	Microbial surface molecules	Opsonization; complement

Abbas p.64–89

3. Toll-like receptors

Humans have 10 functional Toll-like receptors (TLRs), TLR1 to TLR10. Abbas p.65 The cytoplasmic tail of each TLR contains a Toll/IL-1 receptor (TIR) domain. The same domain is present in the receptors for IL-1, IL-18 and IL-33, so these cytokines use similar signalling. Abbas p.67 The function of TLR10 is poorly understood; it may be anti-inflammatory. Abbas p.67

3.1 Ligand and location of each TLR

TLR	Location	Ligand (source)	Adaptor
TLR1:TLR2	Cell surface	Triacyl lipopeptides (bacteria)	MyD88 + MAL
TLR2:TLR6	Cell surface	Diacyl lipopeptides; LTA (gram-positive)	MyD88 + MAL
TLR2	Cell surface	Peptidoglycan, LTA (gram-positive)	MyD88 + MAL
TLR4 (+ MD2, CD14)	Cell surface	LPS (gram-negative); also heat shock proteins (DAMP)	MyD88 + MAL **and** TRIF + TRAM
TLR5	Cell surface	Flagellin (motile bacteria)	MyD88 only
TLR3	Endosome	dsRNA (viruses)	TRIF only
TLR7, TLR8	Endosome	ssRNA (viruses)	MyD88
TLR9	Endosome	Unmethylated CpG DNA (bacteria, herpesviruses)	MyD88
TLR10	—	Unknown; possibly anti-inflammatory	—

Abbas p.67–69 Janeway p.86–90

TLR1, TLR2, TLR4, TLR5 and TLR6 are on the plasma membrane. They recognize bacterial and fungal PAMPs outside the cell. Abbas p.69 TLR3, TLR7, TLR8 and TLR9 are mainly on endosomal membranes and detect nucleic acids. Abbas p.69 Host RNA is not normally present in endosomes. Thus the endosomal location lets these TLRs tell microbial nucleic acids from self nucleic acids. Abbas p.69 The protein UNC93B moves these TLRs from the endoplasmic reticulum to the endosome. Abbas p.69

3.2 TLR9 and CpG DNA

TLR9 binds unmethylated CpG motifs. These motifs are common in prokaryotic DNA but rare in vertebrate genomes. Abbas p.67 In mammalian DNA, methyltransferases heavily methylate the cytosine of CpG dinucleotides. In bacteria and many viruses, CpG stays unmethylated and so acts as a PAMP. Janeway p.88 TLR9 is expressed by plasmacytoid dendritic cells (pDCs), B cells, eosinophils and basophils. Janeway p.86 TLR9-deficient pDCs cannot make type I IFN in response to herpes simplex virus. Janeway p.112

> **Exam pearl**
> "Unmethylated CpG" → TLR9 → endosome. "Methylated CpG" is the host pattern and does not activate TLR9. TLR9 signals through MyD88 to IRF7, so pDCs answer CpG DNA with IFN-α. Abbas p.67 · Janeway p.88, 92

> **Not covered in provided sources — needs expert input**
> The therapeutic use of CpG / immunostimulatory sequences (for example as adjuvants coupled to allergens in allergen immunotherapy) is not described in the provided sources. Add from a current reference if the exam syllabus requires it.

3.3 How TLR4 sees LPS

1. LPS binds soluble LPS-binding protein in blood or tissue fluid. Abbas p.69
2. CD14 is also required for efficient LPS-induced signalling. Abbas p.69
3. The accessory protein MD2 binds the lipid A part of LPS. Abbas p.69
4. The LPS–MD2 complex engages TLR4 and starts signalling. Abbas p.69

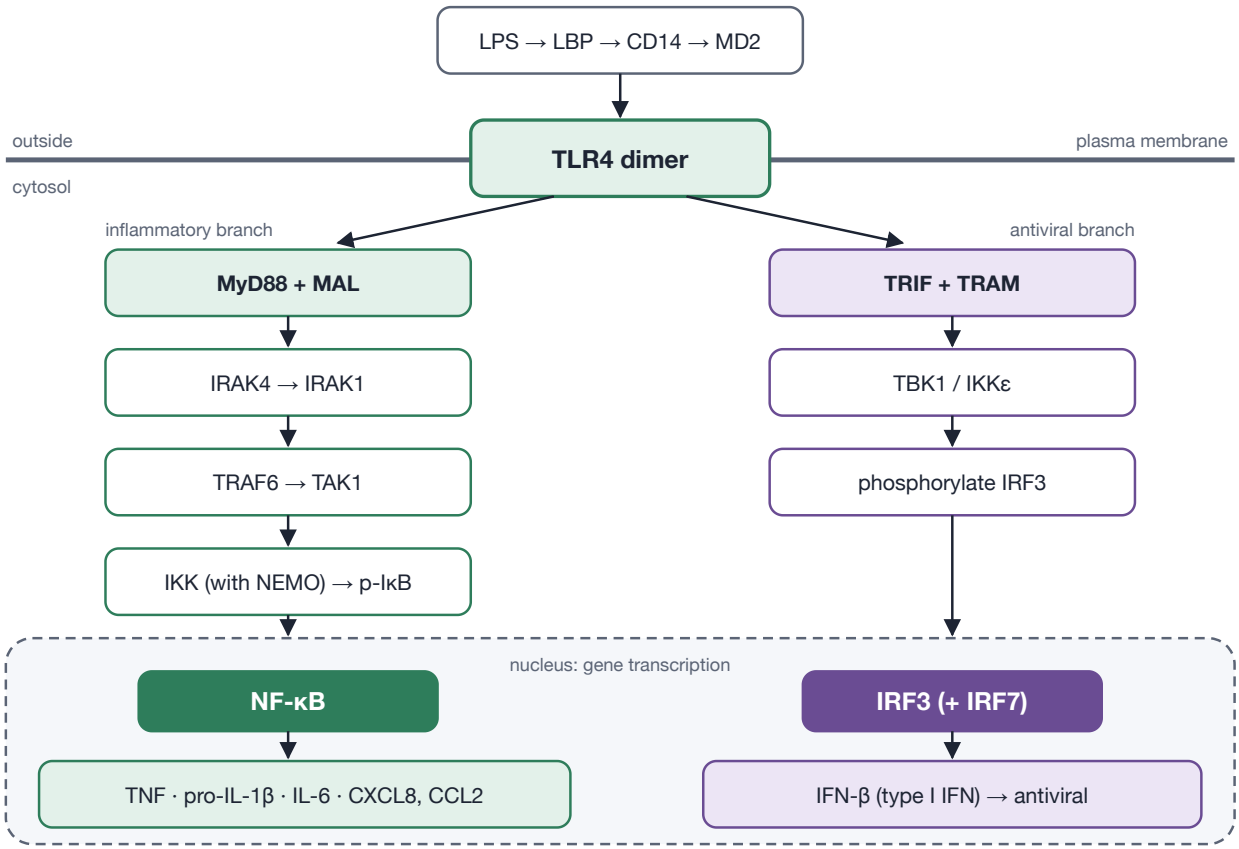
TLRs also respond to DAMPs. For example, heat shock proteins bind TLR4 and HMGB1 binds TLR2. Abbas p.67

4. TLR signalling: MyD88 → NF-κB versus TRIF → IRF3

Ligand binding causes TLR dimerization. The TIR domains then recruit TIR-containing adaptors. Mammalian TLRs use four adaptors: MyD88, MAL, TRIF and TRAM. Janeway p.90 All TLRs except TLR3 use MyD88. Only TLR3 and TLR4 use TRIF. Abbas p.69 TLR4 is the only TLR that uses both adaptor pairs, MyD88/MAL and TRIF/TRAM. Janeway p.90

> **Terminology**
> Janeway uses the name MAL for the adaptor that pairs with MyD88. The alternative name TIRAP, used in the module plan, does not appear in the provided sources — needs expert input to confirm the synonym.

TLR4: two adaptors, two outputs



D8. TLR4 is the TLR that uses both adaptor pairs. MyD88/MAL drives NF-κB and inflammatory genes; TRIF/TRAM drives IRF3 and IFN-β. TLR3 uses only the right branch; most other TLRs use only the left branch.

Abbas p.69 · Janeway p.90–92

4.1 MyD88 pathway to NF-κB (steps)

1. The TIR domain of MyD88 binds the TIR domain of the TLR. Janeway p.91
2. The MyD88 death domain recruits and activates the kinases IRAK4 and IRAK1. Janeway p.91
3. The IRAK complex recruits TRAF6, which leads to activation of the kinase TAK1. Janeway p.91
4. TAK1 activates the IKK complex (IKKα, IKKβ and IKKγ, also called NEMO). Janeway p.91
5. IKK phosphorylates IκB. IκB releases NF-κB, which moves into the nucleus. Janeway p.91
6. NF-κB induces TNF, IL-1, IL-6, chemokines (CCL2, CXCL8) and E-selectin. Abbas p.69 · Janeway p.91

TLR activation also induces the co-stimulatory molecules CD80 and CD86 on dendritic cells and macrophages. These molecules are necessary to start the adaptive immune response. Janeway p.96

4.2 TRIF and endosomal pathways to type I IFN

IRF3 and IRF7, together with NF-κB, induce the type I interferons IFN-α and IFN-β. Abbas p.69 TLR3 signals only through TRIF, which activates NF-κB and IRF3. Abbas p.69 TRIF activates the kinases TBK1 and IKKε. These kinases phosphorylate IRF3, which induces IFN-β. Janeway p.92 Cell-surface TLR4 uses both MyD88 and TRIF, so it gives both inflammatory and antiviral responses. Abbas p.69

The endosomal TLR7, TLR8 and TLR9 also use MyD88, but here MyD88 signalling activates IRFs and type I IFN. Abbas p.69 In endosomes, MyD88 binds IRF7 directly. Thus TLR7 and TLR9 make pDCs produce IFN-α and IFN-β. Janeway p.92

> **Exam pearl**
> "Every TLR except TLR3 uses MyD88." "TLR3 and TLR4 use TRIF." "TLR4 uses both." These three sentences answer most adaptor questions. Abbas p.69 · Janeway p.90

5. NOD-like receptors: NOD1 and NOD2

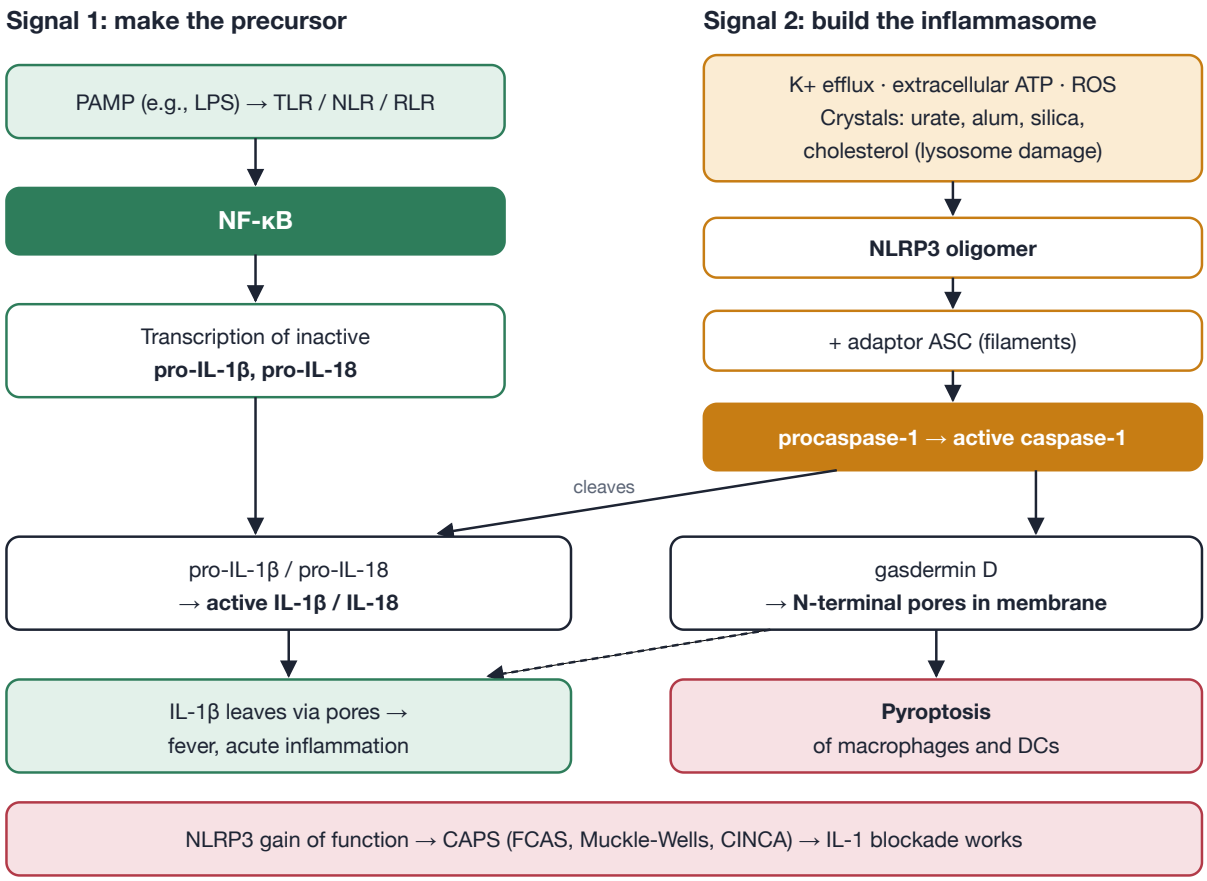
NOD1 and NOD2 are cytosolic NOD-like receptors (NLRs) that sense fragments of bacterial cell wall peptidoglycan. NOD1 recognizes diaminopimelic acid (DAP), mainly from gram-negative bacteria. Abbas p.72 NOD2 recognizes muramyl dipeptide (MDP) from both gram-negative and gram-positive bacteria. Abbas p.72 Activated NODs recruit the kinase RIP2, which activates NF-κB. Abbas p.72

> **Clinical correlation**
>
> Some NOD2 polymorphisms increase the risk of Crohn's disease, probably because the variants sense intestinal microbes poorly. Abbas p.72
>
> Gain-of-function NOD2 mutations cause Blau syndrome. Abbas p.72 Blau syndrome gives spontaneous inflammation of the joints, eyes and skin. Janeway p.93

6. The NLRP3 inflammasome

Inflammasomes are multiprotein complexes. They form in the cytosol in response to infection or cell injury and generate active caspase-1. Abbas p.72 The main function of caspase-1 is to cut the inactive precursors of IL-1β and IL-18 into active cytokines. Abbas p.72

IL-1β production needs two signals. Signal 1 comes from TLR, NLR or RLR signalling through NF-κB and makes the precursor pro-IL-1β. Signal 2 activates the inflammasome, which cleaves the precursor. Abbas p.93



D9. NLRP3 inflammasome in two steps. Signal 1 (NF-κB) supplies inactive pro-IL-1β; signal 2 assembles NLRP3, ASC and caspase-1. Caspase-1 activates IL-1β/IL-18 and cleaves gasdermin D, whose pores release IL-1β and can cause pyroptosis. Abbas p.72–74, 93

6.1 Assembly and output (steps)

1. A trigger changes the cytosol. Common triggers are reduced cytosolic K+ (K+ efflux) and reactive oxygen species. Crystals may act by damaging lysosomal membranes. Abbas p.74
2. NLRP3 molecules oligomerize. Each NLRP3 binds the adaptor ASC. Abbas p.72
3. ASC forms filaments that recruit procaspase-1. Clustering converts it to active caspase-1. Abbas p.72
4. Caspase-1 cleaves pro-IL-1β and pro-IL-18 into active cytokines. Abbas p.72
5. Caspase-1 also cleaves gasdermin D. The N-terminal fragment forms membrane pores. IL-1β lacks a signal peptide and leaves the cell through these pores. Abbas p.72

NLRP3 senses uric acid crystals, alum, cholesterol crystals, extracellular ATP, silica, some streptococcal and staphylococcal toxins and influenza virus. Abbas p.74 The adjuvant effect of alum depends on NLRP3. Janeway p.94 Other inflammasome sensors are NLRC4 (cytosolic flagellin), NLRP1 (anthrax lethal toxin), AIM2 (dsDNA) and pyrin. Abbas p.72–74

6.2 Pyroptosis and the noncanonical pathway

Pyroptosis is an inflammatory programmed cell death of macrophages and dendritic cells. It does not occur in neutrophils. Gasdermin D pores let fluid in; the cell swells and its plasma membrane fails. Abbas p.74 In the noncanonical pathway, cytosolic LPS activates caspase-4 or caspase-5 in humans (caspase-11 in rodents). These caspases cleave gasdermin D and cause pyroptosis. Abbas p.74

6.3 Gout and the inflammasomopathies

Phagocytosed monosodium urate crystals damage lysosomes and activate inflammasomes. IL-1 antagonists are used for severe gout that does not respond to conventional anti-inflammatory drugs. Abbas p.74

Gain-of-function mutations in inflammasome components, or loss-of-function mutations in their regulators, cause excess IL-1. The result is recurrent fever with inflammation of skin, joints and abdomen. These disorders are inflammasomopathies, a subgroup of the autoinflammatory syndromes. Abbas p.74 Autoinflammatory diseases are different from autoimmune diseases, which self-reactive antibodies or T cells cause. Abbas p.74

Disease	Gene (protein)	Key features
CAPS: familial cold autoinflammatory syndrome (FCAS)	*NLRP3* (cryopyrin), dominant	Fever induced by cold, urticarial rash, joint pain, conjunctivitis
CAPS: Muckle-Wells syndrome	*NLRP3*, dominant	Periodic fever, urticarial rash, joint pain, conjunctivitis, progressive deafness
CAPS: chronic infantile neurologic cutaneous and articular syndrome (CINCA)	*NLRP3*	Neonatal-onset fever, urticarial rash, chronic arthropathy, neurological disease
Familial Mediterranean fever (FMF)	*MEFV* (pyrin)	Fever, serositis, arthritis
TNF receptor-associated periodic syndrome (TRAPS)	*TNFRSF1A* (TNF receptor I)	Periodic fever, myalgia, rash; responds to TNF blockade
Hyper-IgD syndrome (HIDS)	*MVK* (mevalonate kinase)	Fever from infancy, high serum IgD, lymphadenopathy
Blau syndrome	*NOD2*, gain of function	Spontaneous inflammation of joints, eyes and skin

Abbas p.74–75 Janeway p.93, 536–537

Mutations in NLRP3 (cryopyrin) cause the cryopyrin-associated periodic syndromes (CAPS). Abbas p.74 CAPS gives fever, neutrophilic rash, bony lesions and central nervous system inflammation. Abbas p.75 Muckle-Wells syndrome and FCAS are dominantly inherited NLRP3 disorders; in FCAS, cold exposure triggers the fever. Janeway p.537

Volatile — check current guidance

Patients with CAPS can be treated successfully with IL-1 antagonists. Abbas p.74 Muckle-Wells syndrome responds dramatically to anakinra, an IL-1 receptor antagonist. Janeway p.537 TRAPS responds to TNF blockade. Janeway p.536 Check drug choice and licensing against current guidance.

Source conflict

Abbas (Table 4.5) lists the *MEFV* defect in FMF as "gain of function, dysregulated inflammasome activation". Janeway (8th ed.) describes an absence of functional pyrin and autosomal recessive inheritance. Follow Abbas (newer) for mechanism questions. Abbas p.75 Janeway p.536–537

7. Cytosolic sensing of viral RNA and DNA

7.1 RIG-like receptors and MAVS

RIG-like receptors (RLRs) are cytosolic sensors of viral RNA that induce type I IFN. The best known RLRs are RIG-I and MDA5. Abbas p.75 RIG-I recognizes RNA with a 5′-triphosphate. Host cytosolic mRNA has a 7-methylguanosine cap and does not carry this signal. Abbas p.75–76 MDA5 recognizes long dsRNA (1–6 kb). Abbas p.75 RIG-I-deficient mice are susceptible to several ssRNA viruses but not to picornaviruses; MDA5-deficient mice are susceptible to picornaviruses. Janeway p.95

1. RIG-I or MDA5 binds viral RNA in the cytosol. Abbas p.75
2. The receptor binds MAVS on the outer mitochondrial membrane. MAVS polymerizes and recruits TRAF proteins. Abbas p.76
3. TRAFs activate TBK1 and IKK. Abbas p.76
4. TBK1 activates IRF3 and IRF7; IKK activates NF-κB. Together they induce type I IFN. Abbas p.76

Exam pearl

The SARS-CoV-2 mRNA vaccines contain pseudouridine instead of uridine. This change reduces detection by innate RNA sensors. Abbas p.76

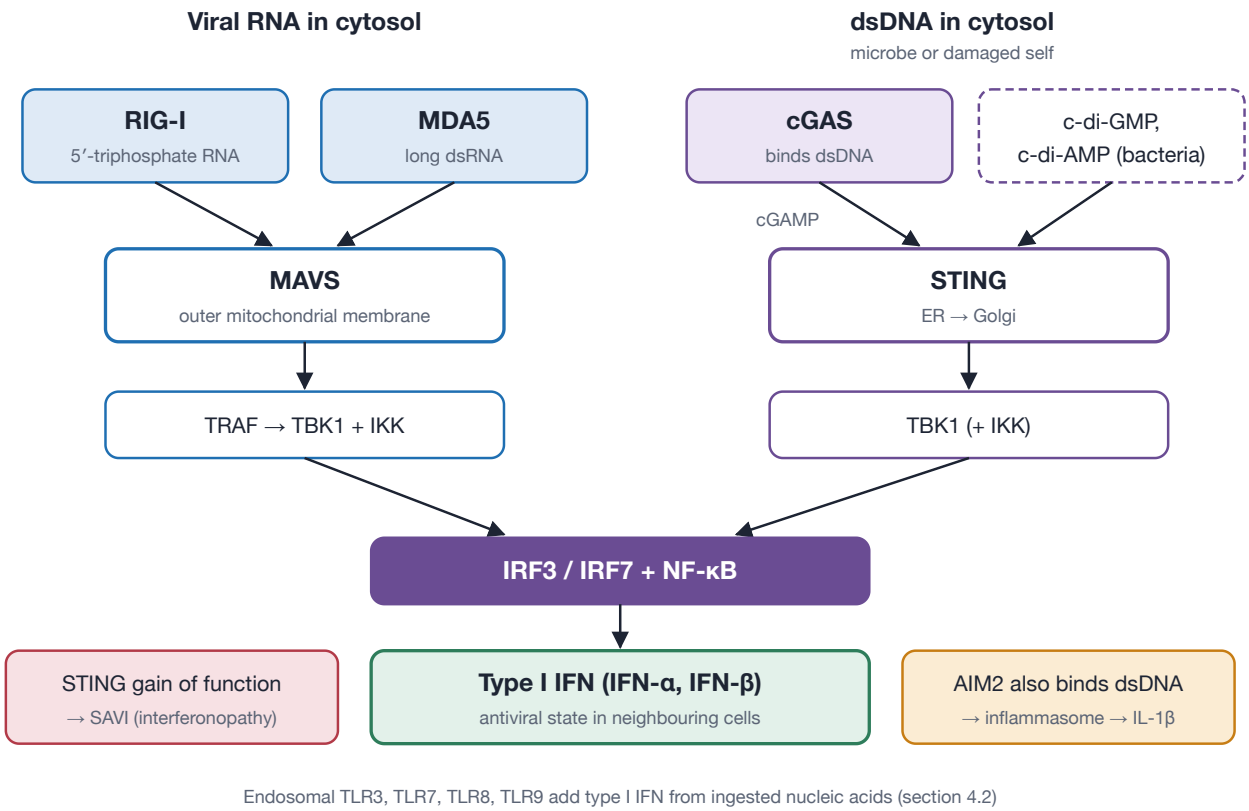
7.2 Cytosolic DNA sensors: cGAS–STING and AIM2

Cytosolic DNA sensors detect dsDNA from microbes or from damaged host cells (micronuclei, mitochondria). They induce type I IFN and autophagy. Abbas p.76

1. Cytosolic dsDNA activates the enzyme cGAS. Abbas p.76
2. cGAS makes the second messenger cyclic GMP-AMP (cGAMP). Abbas p.76
3. cGAMP binds STING in the endoplasmic reticulum. STING moves to the Golgi. Abbas p.76
4. STING activates TBK1. TBK1 activates IRF3 and IRF7, and NF-κB is also activated. Abbas p.76
5. IRFs and NF-κB induce type I IFN genes. Abbas p.76

STING also senses the bacterial second messengers cyclic di-GMP and cyclic di-AMP directly. Abbas p.77

AIM2 is a different cytosolic DNA sensor. It binds dsDNA and forms an inflammasome that generates IL-1β. Abbas p.78



D10. Cytosolic nucleic-acid sensing. RNA sensors (RIG-I, MDA5) signal through MAVS; DNA sensing by cGAS signals through cGAMP and STING. Both routes activate TBK1, IRF3/IRF7 and NF-κB to induce type I IFN. Abbas p.69–97

> **Clinical correlation: interferonopathies**
>
> Gain-of-function STING mutations cause a type I IFN-driven systemic inflammatory disease of skin and lung. Abbas p.77 This is STING-associated vasculopathy with onset in infancy (SAVI): cutaneous vasculopathy, fever and interstitial lung disease. Abbas p.75
>
> Aicardi-Goutières syndrome is caused by loss-of-function mutations in nucleases such as TREX1 and SAMHD1. Excess DNA and RNA then reach the nucleic acid sensors. Abbas p.75

8. Other cell-surface PRRs

8.1 C-type lectin receptors

The mannose receptor (CD206) on macrophages mediates phagocytosis of microbes whose glycans end in mannose, fucose or N-acetylglucosamine. Host cell glycans usually end in galactose and sialic

acid. Abbas p.78 Dectin-1 (CD369) binds β-glucan of fungal cell walls. It is required for immunity to *Candida*, *Aspergillus* and *Pneumocystis*. Abbas p.78 Dectin-1 and Dectin-2 signal through an ITAM–SYK–CARD9 pathway to NF-κB and promote Th17 responses and antifungal immunity. Abbas p.78

8.2 Scavenger receptors and formyl peptide receptors

Scavenger receptors such as SR-A and CD36 on macrophages mediate phagocytosis. CD36 is also a co-receptor for TLR2/6. Abbas p.79 Formyl peptide receptor 1 (FPR1) on leukocytes recognizes peptides that start with N-formylmethionine. Bacteria and mitochondria make such peptides, and FPR1 directs the movement of phagocytes towards them. Abbas p.79

9. Soluble PRRs (brief — lectin pathway in M5)

- **Pentraxins.** Short pentraxins are CRP and serum amyloid P (SAP); the long pentraxin is PTX3. Abbas p.87 CRP binds phosphorylcholine; SAP binds phosphatidylethanolamine. Abbas p.87 All three bind C1q and activate the classical complement pathway. Abbas p.88 Plasma CRP can rise up to 1000-fold in infection, because IL-6, IL-1 and TNF increase hepatic synthesis. Abbas p.88 PTX3 is stored in neutrophil granules and protects mice against *Aspergillus fumigatus* and influenza. Abbas p.88
- **Collectins.** MBL binds terminal mannose and fucose and acts as an opsonin. Low MBL levels increase infection risk, mainly in patients with other immune defects. Abbas p.88 MBL starts the lectin pathway of complement. Abbas p.86 The lung collectins SP-A and SP-D opsonize microbes for alveolar macrophages. Abbas p.88
- **Ficolins.** Ficolins have a fibrinogen-type recognition domain, not a C-type lectin domain. They bind N-acetylglucosamine and LTA, opsonize bacteria and activate complement like MBL. Abbas p.89

10. What PRR signalling produces: innate cytokines

Cytokine	Main effects
TNF	Endothelial and neutrophil activation; acute-phase proteins; fever; cachexia; apoptosis. Main mediator of septic shock.
IL-1 (IL-1β)	Endothelial activation; fever; acute-phase proteins; Th17 differentiation. Needs the inflammasome.
IL-6	Acute-phase reactants in the liver; Th17 differentiation.
IL-12	IFN-γ from ILC1s, NK cells and T cells; Th1 differentiation.
IL-18	Enhances NK cell function; needs the inflammasome.
Type I IFN (IFN-α, IFN-β)	Antiviral state, mainly in uninfected neighbouring cells; more MHC class I; NK cell activation. pDCs are the main IFN-α source.
Chemokines (CXCL8, CCL2)	Recruit neutrophils (CXCL8) and monocytes (CCL2).

Abbas p.90–98

TNF, IL-1 and IL-6 make hepatocytes produce acute-phase proteins such as CRP, SAP and fibrinogen. Abbas p.97 Septic shock is mediated in large part by TNF. Abbas p.92 Patients with IL-12 receptor β1 mutations are highly susceptible to *Salmonella* and mycobacteria. Abbas p.93

11. Inborn errors of PRR signalling

Defect	Pathway hit	Clinical clue
MyD88 or IRAK4 deficiency (autosomal recessive)	MyD88 pathway of most TLRs and the IL-1 receptor	Severe invasive pyogenic infections early in life, especially pneumococcal; little fever or inflammation ("cold" infection); better with age; no extra viral susceptibility reported in IRAK4 deficiency
TLR3, UNC93B, TRIF, TRAF3 or TBK1 defects	TLR3–TRIF → type I IFN	Herpes simplex encephalitis; other viral infections not clearly increased
NEMO (IKKγ) deficiency, X-linked (EDA-ID)	NF-κB downstream of all TLRs (and CD40)	Ectodermal dysplasia plus pyogenic, mycobacterial, viral and *Pneumocystis* infections

Abbas p.489, 498 Janeway p.92, 535

The MyD88–IRAK4–IRAK1 pathway lies downstream of most TLRs and of the IL-1 receptor. Abbas p.489 Patients with MyD88 or IRAK4 mutations have severe invasive bacterial infections early in life, especially pneumococcal pneumonia. Infections become less severe later. Abbas p.489 These patients usually cannot mount a fever during illness. Abbas p.489 The infections cause little inflammation and are called "cold" infections. Janeway p.535 IRAK4 deficiency does not add susceptibility to viral infections. Janeway p.92

Loss-of-function TLR3 mutations cause recurrent herpes simplex encephalitis (HSE). Abbas p.489 Autosomal recessive TRIF mutations and autosomal dominant TRAF3 or TBK1 mutations give the same susceptibility. Abbas p.489 Homozygous UNC93B mutations reduce type I IFN production and also predispose to HSE. Abbas p.489 NEMO deficiency blocks NF-κB downstream of all TLRs, so it affects innate and adaptive immunity. Janeway p.535

> **Exam pearl**
>
> Invasive pneumococcal infection with little fever or local inflammation → think MyD88/IRAK4. Isolated HSE in an otherwise healthy child → think TLR3 pathway. Abbas p.489 · Janeway p.92, 535

12. Links to allergy

Mast cells express TLRs. Abbas p.86 IL-33 is stored in the nuclei of barrier epithelial cells. When damaged cells release it, IL-33 acts as an alarmin. Abbas p.94

Source conflict

Abbas chapter 4 states that TLR ligands can induce mast cell degranulation. Abbas p.86 Abbas chapter 20 lists TLR ligands among stimuli that make mast cells produce lipid mediators, cytokines and chemokines without degranulation. Abbas p.470 Avoid absolute answers on this point.

5 key points

1. Surface TLRs (1, 2, 4, 5, 6) see bacterial walls; endosomal TLRs (3, 7, 8, 9) see nucleic acids; TLR9 sees unmethylated CpG DNA. Abbas p.67–69
2. All TLRs except TLR3 use MyD88 → NF-κB; TLR3 and TLR4 use TRIF → IRF3 → IFN-β. Abbas p.69 · Janeway p.92
3. IL-1β needs two signals: NF-κB makes pro-IL-1β; the NLRP3–ASC–caspase-1 inflammasome cleaves it and gasdermin D pores release it. Abbas p.72, 93
4. Cytosolic RNA (RIG-I, MDA5 → MAVS) and DNA (cGAS → STING) sensing converge on TBK1–IRF3/IRF7 and type I IFN. Abbas p.76
5. MyD88/IRAK4 deficiency → cold pyogenic infections; TLR3-pathway defects → HSE; NLRP3 gain of function → CAPS, treated with IL-1 blockade. Abbas p.74, 489 · Janeway p.535

3 exam traps

1. TLR4 is a surface TLR but still uses TRIF. TLR3 is the only TLR that does not use MyD88. Abbas p.69 · Janeway p.90
2. Pyroptosis affects macrophages and dendritic cells, not neutrophils. Abbas p.74
3. NOD2 variants that sense microbes poorly raise Crohn's risk, but NOD2 gain of function causes Blau syndrome. Do not mix the direction of the mutation. Abbas p.72

Practice: M3 questions in the quiz (IDs M3-A1 … M3-K3).

Sources: Abbas, Cellular and Molecular Immunology, 11th ed., pp. 63–98 (ch. 4), 470 (ch. 20), 487–498 (ch. 21); Janeway's Immunobiology, 8th ed., pp. 86–96, 112 (ch. 3), 535–537 (ch. 13). · Fact IDs refer to `ledger/facts.csv`.